

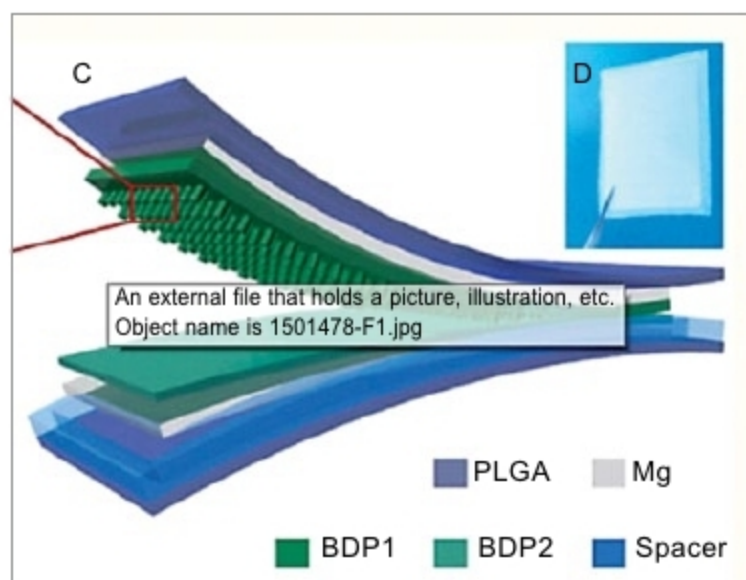


Fig. 6: Multi-layered structure of BD-TENG

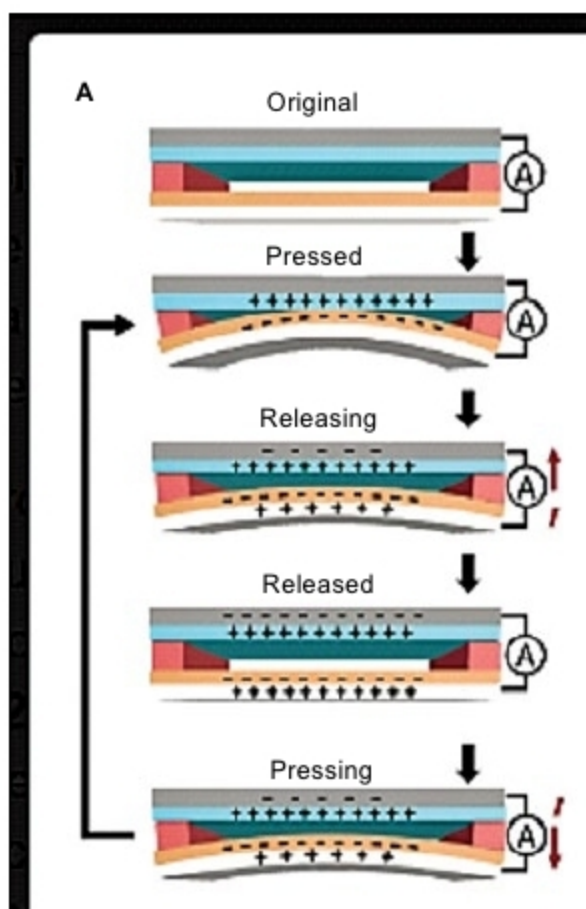


Fig. 7: Operation of BD-TENG

side of the wound and serve to generate an electric field that could penetrate into the dermis, and strengthen the endogenous electric field for enhanced wound healing. Further, the electric field strength outside electrode coverage drops quickly to a negligible value, as compared to the high value in between. Hence, the electric field has minimal impact on the surrounding tissue or organs.

Biodegradable triboelectric nanogenerator

There is another type of medical nanogenerator, that is, biodegradable triboelectric nanogenerator (BD-TENG), which can be implanted to harvest in vivo biomechanical energy, and then degraded and resorbed in the body

without any adverse effect after completing its work cycle. So, no surgery is needed to remove it.

The nanogenerator generates DC pulsed electrical field between two complementary micro-grating magnesium electrodes, which successfully repair damaged cells like neuron and other types of cells. BD-TENG is a multi-layered structure comprising biodegradable polymers and resorbable metals. Open circuit voltage of BD-TENG can reach up to 40V approximately, and the corresponding short-circuit current is one micro-amp. The encapsulated structure consists of friction layers, electrode layers and spacers, as shown in Fig. 6.

When an external force brings two BDP friction layers (polylactide-glycolide, or PLGA, and polycaprolactone, or PCL) into contact, electrons are transferred from PLGA to PCL and retained on both BDPs, resulting in surface triboelectric charges. Subsequent release of the external force causes separation of PLGA from PCL films and forms an electrical potential between these two films that drives electrons through external load and screens inductive charges on both electrodes.

This performance can be tuned from 10V to 40V by changing the composition of polymers. After four weeks of implantation, the device is completely biodegraded. Mg electrodes are hydrolysed. This process is known as vertical separation mode of TENG. PLGA and PCL both are biodegradable polymers.

Conclusion

TENG is a fascinating technology that dramatically speeds up healing. It reduces healing time to one-fourth of the normal healing process. It can heal even nasty wounds of a diabetic patient. Moreover, BD-TENG is biodegradable. This means there is no need to open up the patient's body for taking out the implant. In the near future, this will be a very powerful tool in the hands of doctors. **EFY**

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		0 - 200
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mA	0 - 2, 20, 200	
	A	0 - 2, 5
	AC	V
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A		0 - 1, 5, 20

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MECO INSTRUMENTS PRIVATE LTD.

Plot No. EL-1, MIDC Electronic Zone,
TTC Industrial Area, Mahape,
Navi Mumbai - 400710 (INDIA)

Tel : 0091-22-2767 3300 (Board), 2767 3311-16 (Sales)
Fax : 0091-22-27673310
Email : sales@meconinst.com Web : www.meconinst.com
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